

Overfit & Overcaffeinated

Catching fraud that moves — without ever seeing the future.

Behavioural features



Two horizons + a neural net



Neighbour propagation

0.56548

Public leaderboard

3.4×

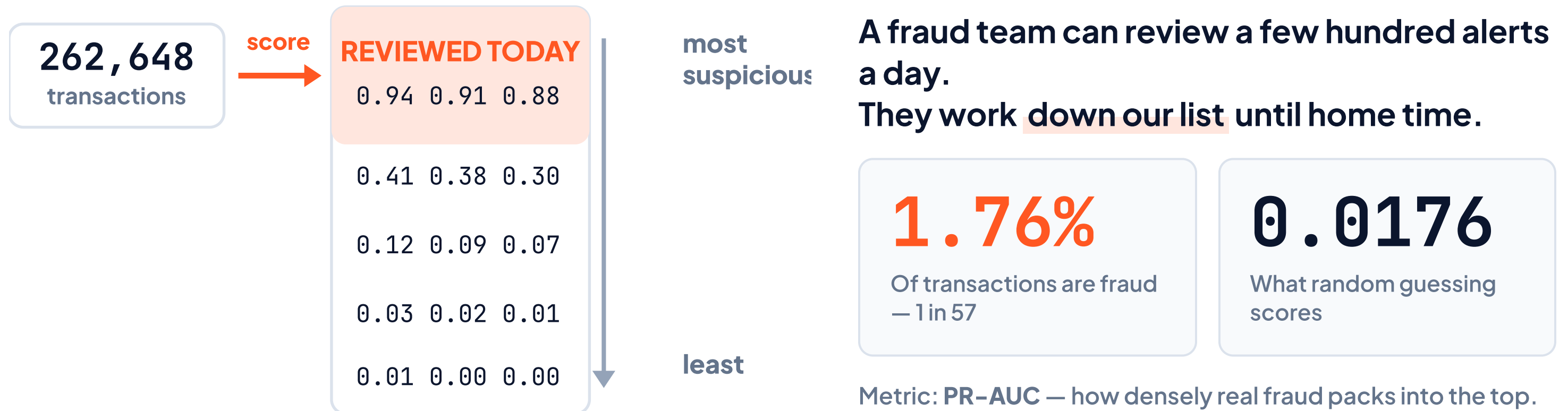
The baseline PR-AUC

0

External data or pretrained models

PROBLEM & FRAMING

We don't catch fraud. We decide **who gets checked first.**



The test period starts **where our data ends.**

TRAIN · 731,942 rows

1 Jan → 15 Jul · labelled

TEST · 262,648 rows

16 Jul → 15 Sep · blind

2 months forward · no labels · no feedback

This is **forecasting**, not pattern-matching.

13

Raw columns. No text, no images,
nothing exotic.

99.8%

Of test rows use a device we already
have history for

5/day

Submission cap · ~40 hours total

PROBLEM & FRAMING

No single transaction here is suspicious.

22:45:58	598.88	restaurant	FRAUD
22:46:04	782.70	restaurant	FRAUD
22:46:59	609.88	restaurant	FRAUD
22:47:05	737.05	education	FRAUD
22:47:11	667.79	financial	FRAUD
22:47:48	517.00	restaurant	FRAUD
22:48:24	340.24	restaurant	FRAUD
22:49:28	691.76	education	FRAUD
22:51:53	540.66	financial	FRAUD
22:52:40	624.29	electronics	FRAUD

Device D000775 · 12 transactions in 7 minutes · all fraud

486 tk

Median legitimate transaction

2,310 tk

Median fraud transaction

This burst runs at 340–780 tk.
Cheaper than typical fraud.
Every amount rule misses it.

The pattern exists only across rows — and that drove the whole architecture.

We stopped describing transactions. We described **behaviour**.

13 raw columns

amount · hour · category · device type



160 behavioural features

every one a comparison against that customer's, device's or merchant's **own past**

6,550 tk

Globally: 98th percentile. 14,600 legitimate transactions are bigger. Useless as a flag.

64× normal

For this customer: who has never once spent above 169 tk.
Now it screams.

Raw columns

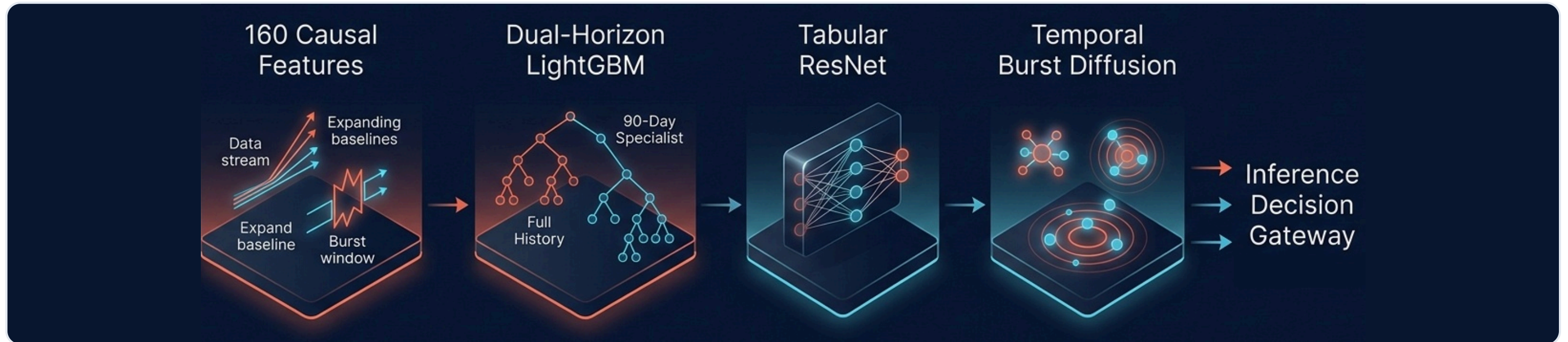
0.1664

+ behavioural

0.5090

+0.34 in one step — more than every change that followed, combined.

The whole solution, **in one pass.**



160 features → 0.48 full-history + 0.52 90-day → 0.88 trees + 0.12 ResNet → ±30 min neighbour pass → **ranked queue**

Why two horizons

One advisor has read six months of case files. One has read only the last ninety days. **The recent one gets the bigger vote.**

Why a neural net at 12%

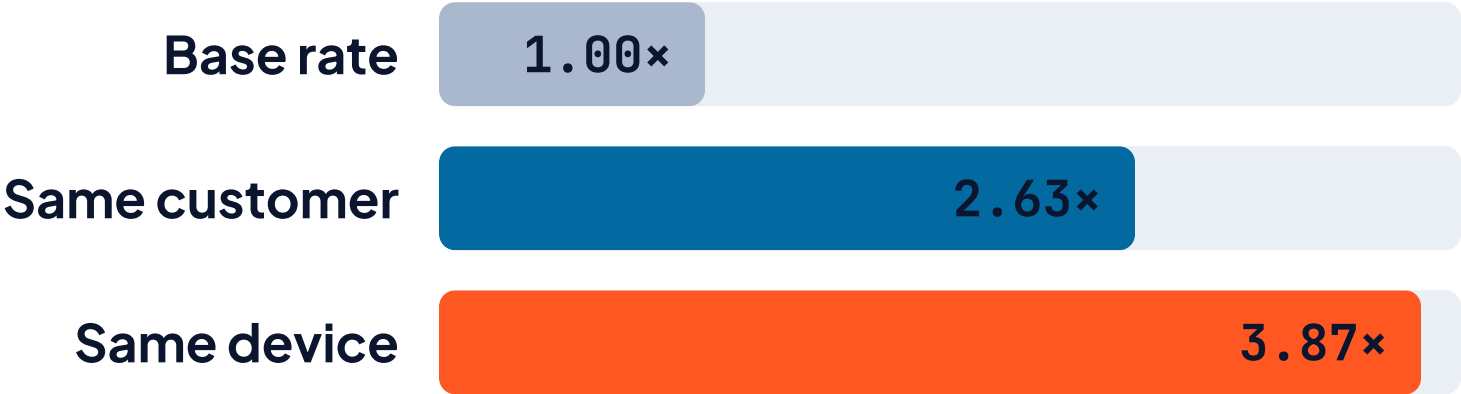
Alone it scores **0.5173** — worse than the trees. It is in because it is wrong about **different rows.**

What it costs

~15 minutes on CPU, end to end. Single seed, every random source pinned. **Zero** pretrained models.

Fraud has **siblings**.

If this transaction is fraud, how likely is its neighbour?



Measured across all 731,942 training transactions.

Half its own score, half its neighbours'

0.50 its own score + **0.50** average of neighbours

±30 min
window, either side

customer
or device
what counts as a
neighbour

excluded
from its own average — it
cannot vouch for itself

Uses **no labels** — only model outputs, entity ids and timestamps.

BEFORE — judged alone



all mildly odd · none flagged

AFTER — neighbours share



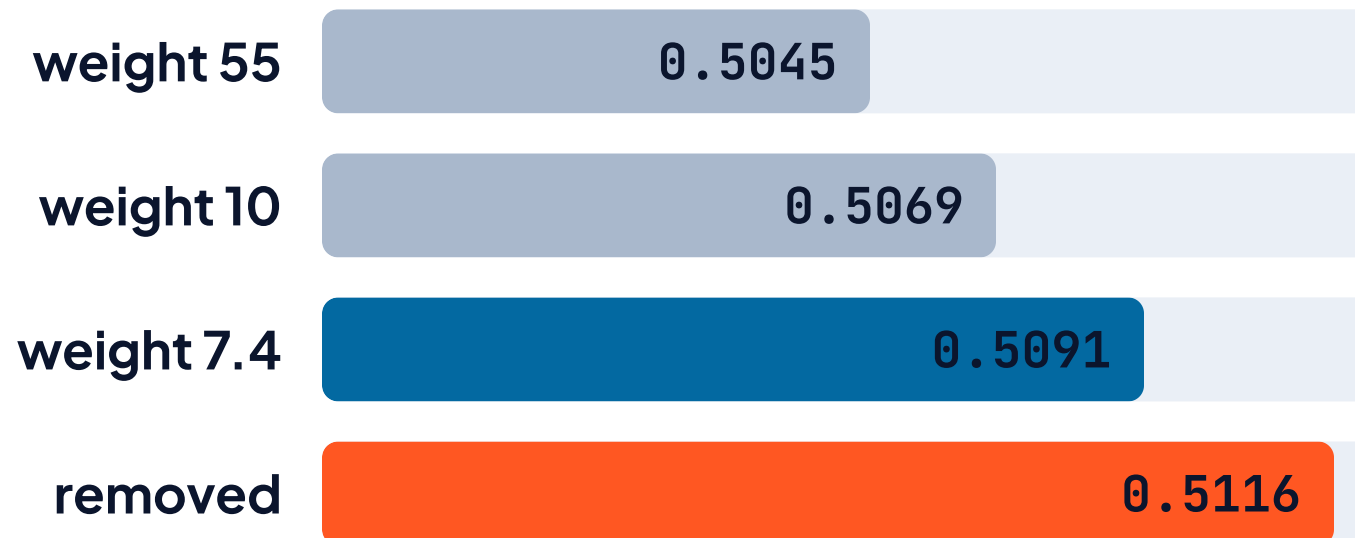
together they clear the bar

95.3% of test rows have no in-window neighbour and pass through **completely unchanged**. This refines the ranking — it doesn't produce it.

KEY TECHNICAL DECISIONS

Our biggest gain was deleting one line.

Class weighting told the model to treat every fraud row as 55 rows.



Every step down helped. No optimum in the middle — that shape is a mechanism, not a lucky run.

Why it hurt

Weighting is for **deciding** — block a card, freeze an account.

We never decide. We **order a queue** — and it made the model flag generously, crowding our top with **false alarms**.

+0.026

On the leaderboard — more than every feature built that day combined

In our plan from round one; every sweep held it fixed. **Every conclusion drawn under it was void.**

KEY TECHNICAL DECISIONS

Seven experiments. All of them noise.

Our scale wobbles by half a kilo.
We tried seven diets, each showed 0.3 kg.
We were reading the wobble.

0.0020

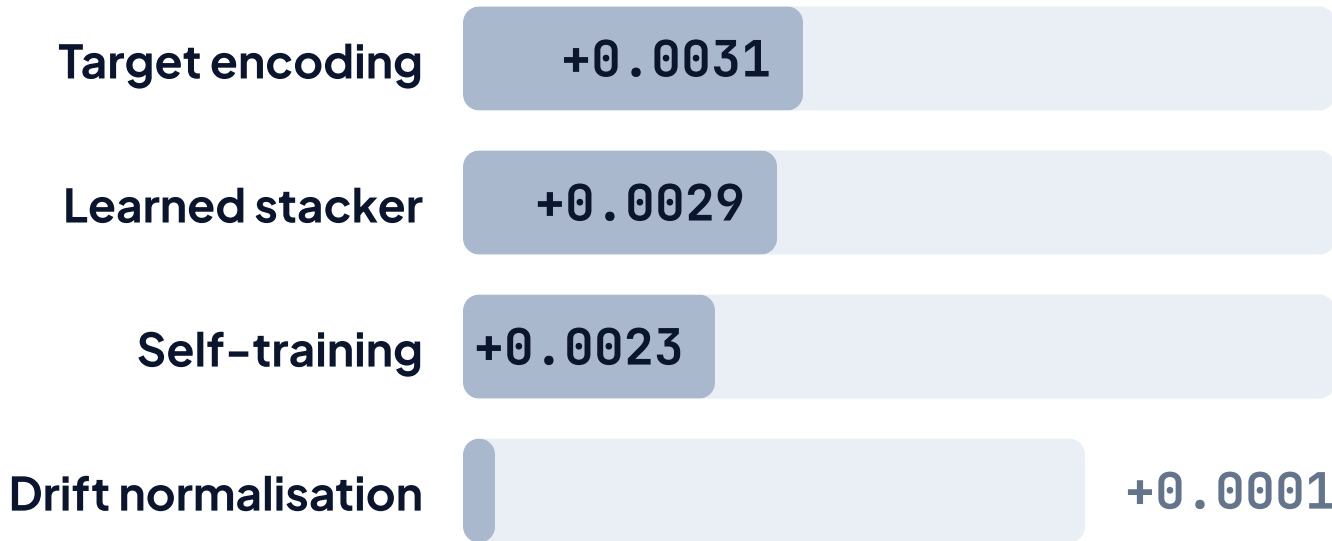
Score movement from changing nothing but the random seed

CatBoost · first sweep · recency weighting · graph features · stationarity fixes · ratio features · expanded windows — **every one landed inside it.**

The rule that followed

Multi-seed mean, and an effect at least 2× the noise floor, or it does not count.

Then rejected — despite all being **positive**:



KEY TECHNICAL DECISIONS

Fraud changed shape in July. Here is **what we did about it**.

Customer-amount signal 0.41 → 0.24 HALVED	Device-velocity signal 0.095 → 0.172 DOUBLED	Fraud rate 1.5 – 1.9% UNCHANGED ALL YEAR
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Fraud didn't get **rarer**. It got **different** — and the test window sits **entirely after** the change.

OUR FIX

A **second model** trained only on the last 90 days — given the **bigger vote**.

48%
full history · 6.5 months

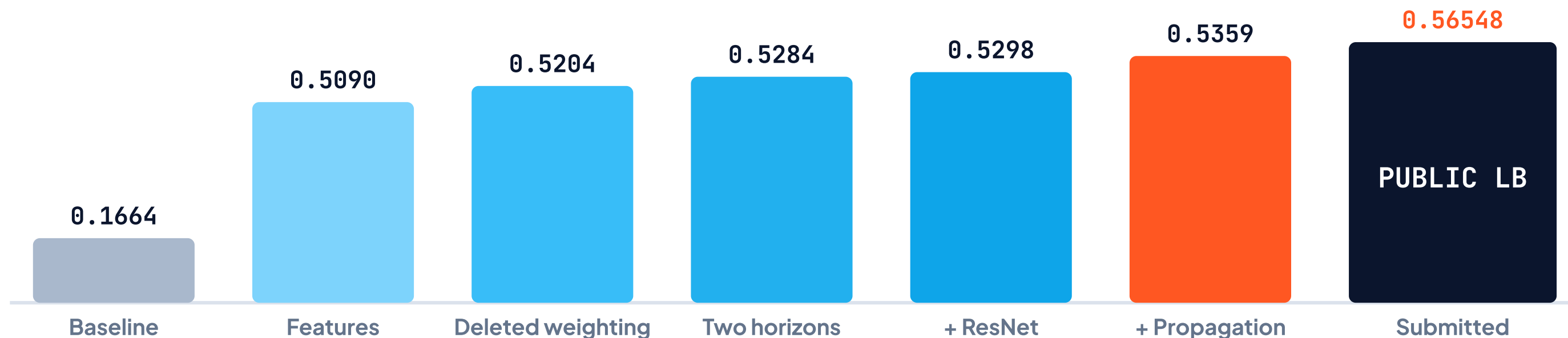
52%
last 90 days only

The less experienced model wins the argument, because everything it knows is **current**.

Tested: drop the decayed features -0.0010 The obvious move, and it made things worse — 0.24 still beats 0.172 . The faded signal was still the strongest thing we had.	Tested: weight recent rows worse at every setting 0.5449 → 0.5401 → 0.5361 → 0.5308. Down-weighting old rows doesn't make the model younger , it just discards signal.
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RESULTS & LEADERBOARD

0.56548 — 3.4× the baseline.



Where it breaks - 1

Lone wolves. A first-ever transaction on a fresh device has no history and no siblings — both our strongest mechanisms are blind to it.

Where it breaks - 2

The model ages. We measured -0.019 over 60 days of training staleness — and the test window runs 62 days past training.

The ceiling is the data

All 33 active teams landed between **0.50** and **0.565**. Nobody found a magic signal, because there wasn't one.

IMPACT & CONCLUSION

You don't review a transaction. You review an **account**.

Auto-block

Top of the queue. Burst attacks on one device, caught as a group rather than one row at a time.

Step-up verification

The rescued middle — transactions that only look suspicious once their siblings are considered.

Pass through

The 95.3% with no suspicious neighbours. Zero friction, no review cost.

Every feature is strictly past-only, so the same design runs in a **live payment pipeline** — not just a leaderboard.

OUR TAKEAWAY

**We won by measuring our own noise
before believing our own results.**

Appendix

Not part of the timed talk — reference only if judges ask.

A · Features

B · Models

C · Compliance

D · Rejected ledger

160 features, three entities, nine horizons.

42

merchant-relative

40

customer-relative

40

device-relative

13

counterparty fan-out

4

network cluster size

5

raw columns kept

Every name reads the same way

dev

amt_vs_recent_max

5m

dev

amt_vs_recent_max

5m

compared against this device's own past — or cust, merch, loc

what is measured — amount, count, money sum, gap, fresh counterparties

the window — 5m to 168h, or _prior for all history

Windows matched to rhythm

device

28 txns each

customer

10

merchant

10

Devices are 3× busier, so devices get 5-minute windows and customers get 30. A 5-minute customer window would be empty almost always — a dead column crowding out live ones.

Nine horizons

5m · 15m · 30m

1h · 3h · 6h · 12h

24h · 72h · 168h

The 5-minute window is what sees twelve transactions in seven minutes.

14

Exact configuration.

Full-history trees

882 rounds

127 leaves

lr 0.02

feature frac 0.85

min leaf 50

All 731,942 rows. 48% of the tree blend.

90-day specialist

433 rounds

63 leaves

lr 0.02

feature frac 0.75

min leaf 100

Recent data only, shallower and more constrained. 52% — the bigger vote.

Tabular ResNet

3 residual blocks

hidden 256

dropout 0.15

AdamW lr 1e-3

8 epochs, cosine

Embeds attributes, never entity IDs. Alone: 0.5173.

No sample weighting

Recency weighting failed monotonically:

none 0.5449

60-day 0.5401

30-day 0.5361

14-day 0.5334

7-day 0.5308

Down-weighting old rows doesn't make the model younger.

160 features → 0.48 full + 0.52 90-day → 0.88 trees + 0.12 ResNet → ±30 min neighbour pass → **ranked queue**

Single seed (42), every random source pinned · zero pretrained models · ~15 minutes end to end on CPU

The pipeline **refuses to run** if a rule is broken.

The rules

1. Only data **before t**
2. Test rows as history — **never the label**
3. **No random K-fold**
4. Behaviour, **not entity IDs**

How they hold

1. Windows are **half-open**
2. Totals **subtract the current row**
3. Cluster size read **before** linking

Three assertions, all called

1. First transaction must be **blank**
2. No forward-looking names
3. **No raw IDs, no label**

Built, then deleted

Forward-looking features scored **+0.024** — our largest single gain.

Against the rules.
Deleted, unsubmitted, now **blocked by assertion**.

One disclosed judgement call: the neighbour window is **symmetric**, so it reads rows after *t*. Stated in writing · changes **4.7%** of rows · a strictly-past submission scoring **0.55368** was entered alongside.

Everything measured – including what we rejected.

MILESTONE	HOLDOUT	PUBLIC LB
Baseline — raw columns	0.1664	0.1664
Behavioural features	0.5090	0.5105
Removed class weighting	0.5204	0.5395
Pruned to top 160	0.5269	—
Dual horizon	0.5284	—
+ Tabular ResNet	0.5298	—
+ Propagation	0.5359	0.56548

Dashes: verified locally without spending a leaderboard submission — we only had 5 per day.

Rejected — all positive, all discarded

Target encoding	+0.0031
Learned stacker	+0.0029
Self-training	+0.0023
Drift normalisation	+0.0001

Noise floor 0.0020 — measured by re-running an identical model under different seeds. Nothing counted unless it cleared **twice** that.

Also dead: CatBoost · ranking objectives (-0.009) · isotonic calibration · recency weighting · graph features (net-neutral) · dropping decayed features (-0.0010).